Outline of the Lecture

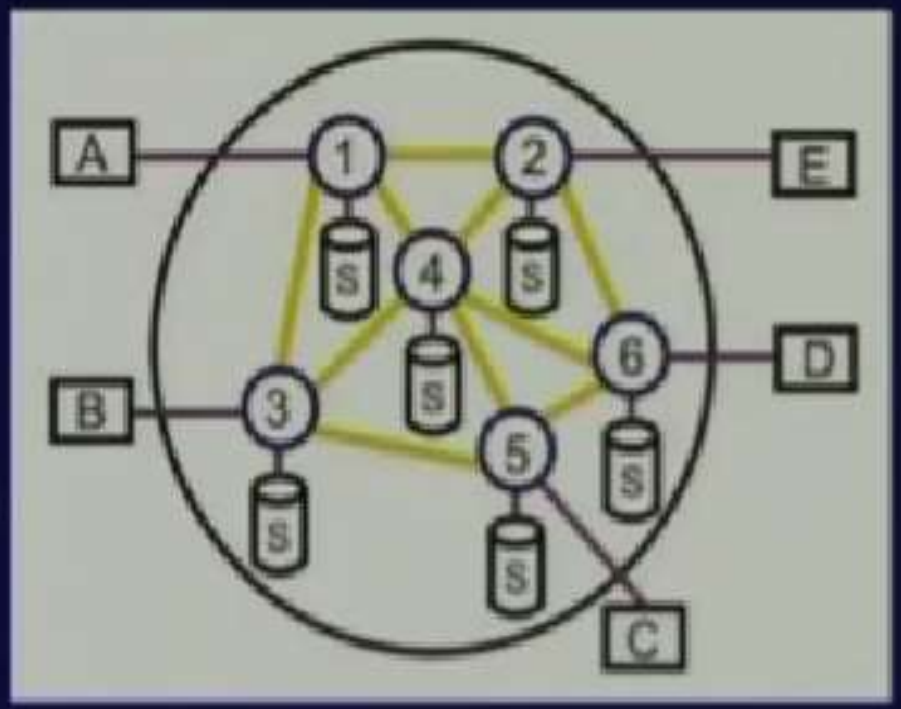
- Routing defined
- Desirable properties of routing
- Design parameters
- Routing algorithms
  - Fixed (Static) routing
    - Dijkstra's Shortest-path algorithm
  - Flooding

Routing Defined

- In most of the situations, packets will require multiple hops to make journey towards the destination station.

Station A
to
station A → D

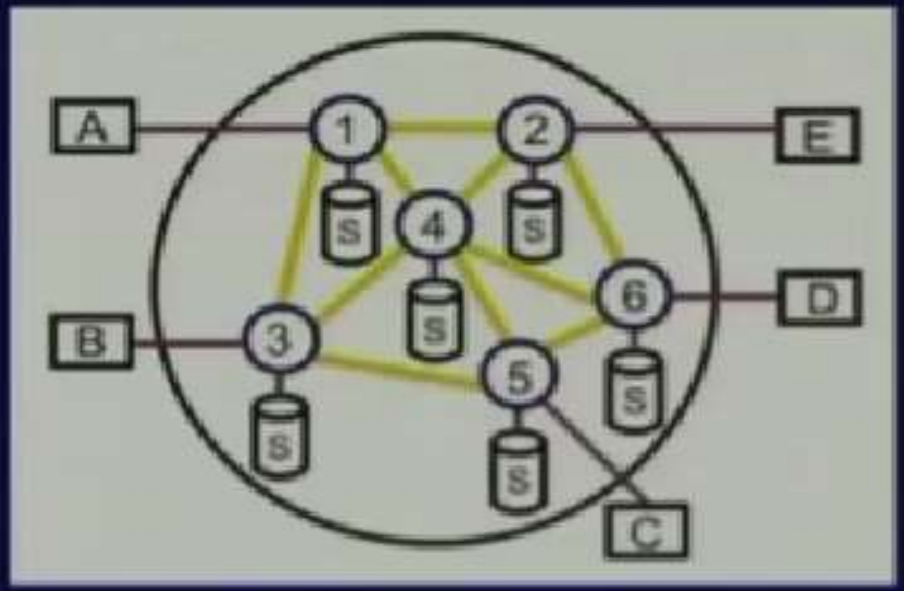
1-2-6
1-3-5-6
1-4-6



Routing Defined

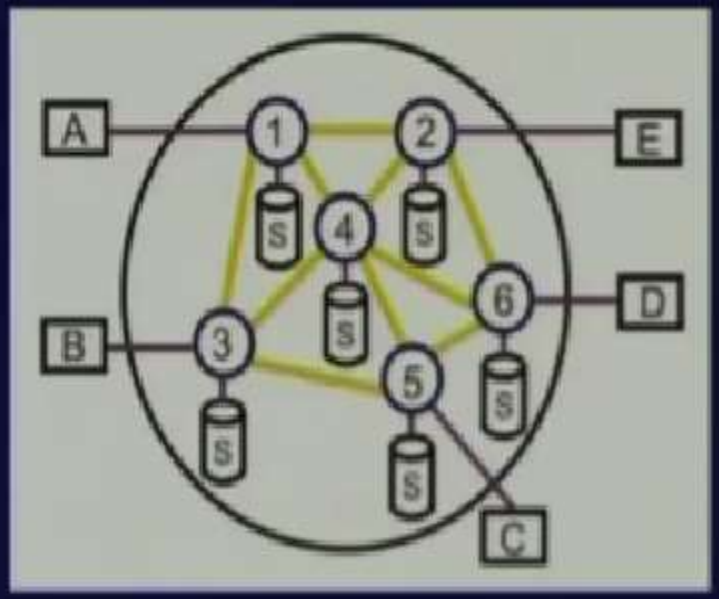
- In most of the situations, packets will require multiple hops to make journey towards the destination station.

•Routing is one of the most complex and crucial aspect of packet-switched network design.



Desirable properties

- Correctness and simplicity: Self-explanatory.



Desirable properties

- **Correctness and simplicity:** Self-explanatory.
- **Robustness:** Ability of the network to deliver packets via some route even in the face of failures.
- **Stability:** The algorithm should converge to equilibrium fast in the face of changing conditions in the network.
- **Fairness and optimality:** obvious requirements, but conflicting.

Waitage
Priority

- **Efficiency:** Minimum overhead

Design Parameters

- **Performance Criteria:** Number of hops, Cost, Delay, Throughput
- **Decision Time:** Per packet (Datagram), per session (Virtual-circuit)
- **Decision Place:** Each node (distributed), Central node (centralized), Originated node (source)
- **Network Information Source:** None, Local, Adjacent node, Nodes along route, All nodes

- **Network Information Update Timing:** Continuous, Periodic, Major load change, Topology change

Routing Strategies

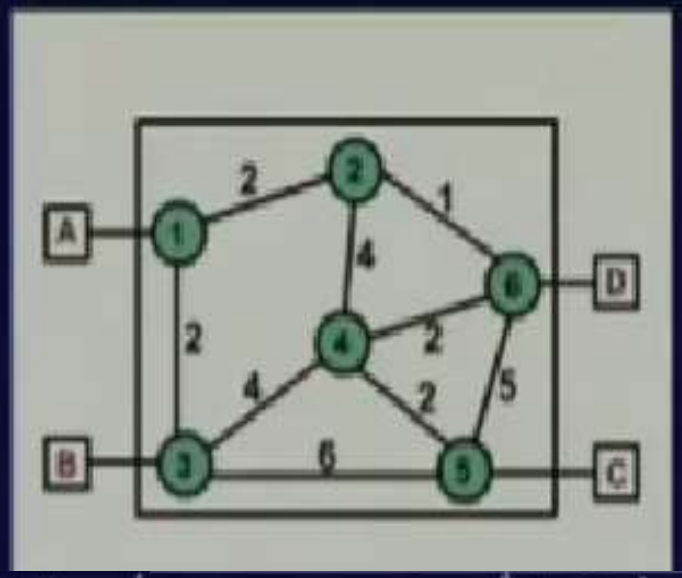
- A large number of routing strategies have evolved over the years.
- Important strategies are:
  - Fixed (Static) routing
  - Flooding
  - Random routing
  - Flow-based Routing
  - Adaptive/Dynamic routing

Fixed Routing

- A route is selected for each source-destination pair of nodes in the network.
- The routes are fixed; they may only change if there is a change in the topology of the network.
- How fixed routing may be implemented?

Fixed Routing: Example

- A central routing matrix is created based on least-cost path, which is stored at a network control center.
- The matrix shows, for each source-destination par of nodes, the identity of the next node on the route.



Central Routing Directory

From node \ To node	1	2	3	4	5	6
1	-	2	3	2	2	2
2	1	-	1	6	6	6
3	1	1	-	4	5	1
4	6	6	3	-	5	6
5	4	4	3	4	-	4
6	2	2	2	4	4	-

Not very reliable

Fixed Routing: Example

- From the main routing matrix, routing tables to be used by each individual node can be developed

Node 1 Directory

Destination	Next Node
2	2
3	3
4	2
5	2
6	2

Node 2 Directory

Destination	Next Node
1	1
3	1
4	6
5	6
6	6

Node 3 Directory

Destination	Next Node
1	1
2	1
4	4
5	5
6	1

Node 4 Directory

Destination	Next Node
1	6
2	6
3	3
5	5
6	6

Node 5 Directory

Destination	Next Node
1	4
2	4
3	3
4	4
6	4

Node 6 Directory

Destination	Next Node
1	2
2	2
3	2
4	4
5	4

Distributed

Least-cost Path

- A cost is associated with each link.
- The simplest criterion is to choose the minimum-hop route through the network.
  - Easily measured criterion.
- A generalization is least-cost routing.
- For any pair of attached stations, the least cost route through the network is looked for.
- For either case, several well-known algorithms exist to obtain the optimum path.
  - Dijkstra's algorithm
  - Bellman-Ford algorithm

Dijkstra's Algorithm

- It finds the shortest paths from a given source node to all other nodes in order of increasing path length
- The algorithm converges under static conditions of topology and link cost
- There are basic steps:
  - Initialization; $M = \{s\}$, $D_n = d_{sn}$ for $n=s$
  - Find neighboring nodes not in M that has least-cost path from s and include in M.
  - Update least cost paths

Dijkstra's Algorithm

Iteration	M	C(2) path	C(3) path	C(4) path	C(5) path	C(6) path
1	{1}	2, 1-2	2, 1-3	∞	∞ –	∞ –
2	{1, 2}	2, 1-2	2, 1-3	6, 1-2-4	∞ –	3, 1-2-6
3	{1, 2, 3}	2, 1-2	2, 1-3	5, 1-2-6-4	7,1-2-6-4-5	3, 1-2-6
4	{1, 2, 3, 6}	2, 1-2	2, 1-3	5, 1-2-6-4	7,1-2-6-4-5	3, 1-2-6
5	{1,2,3,4,6}	2, 1-2	2, 1-3	5, 1-2-6-4	7,1-2-6-4-5	3, 1-2-6
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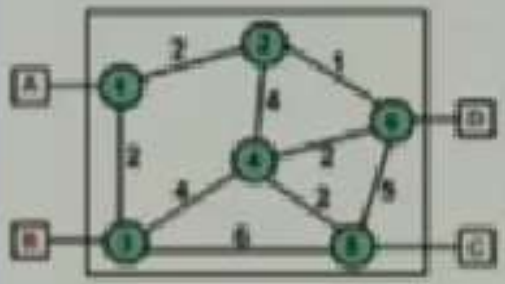
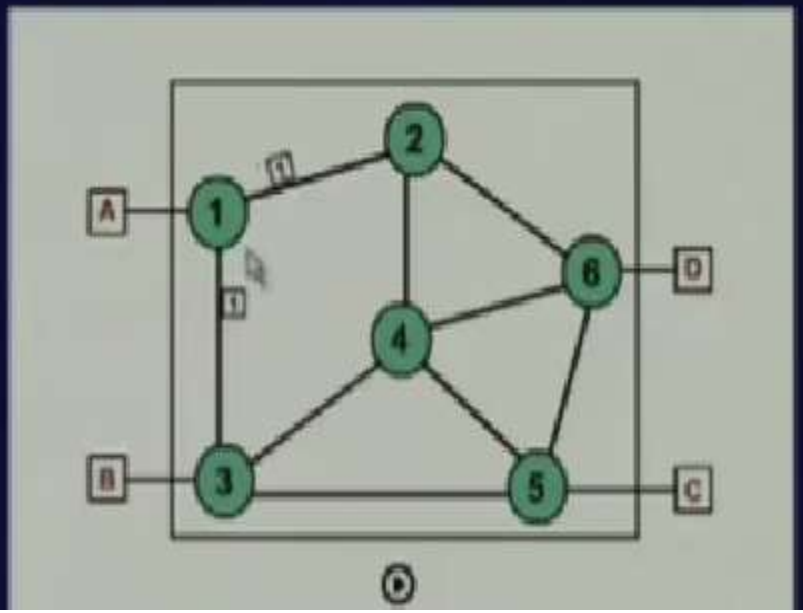


Fixed Routing

- Advantages:
  - Simple
  - Works well in a reliable network with stable load ~~in a reliable network~~
  - Same for virtual-circuit and Datagram
- Disadvantage:
  - Lack of flexibility.
  - Does not react to failure or network congestion

Flooding

- Requires no network information whatsoever
- Every incoming packet to a node is sent out on every outgoing line except the one it arrived on.



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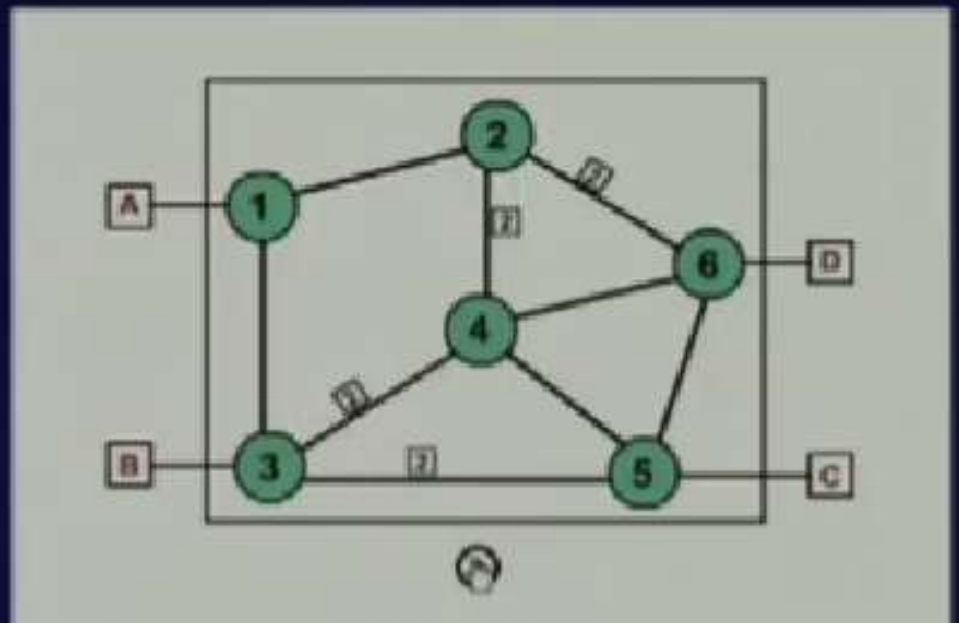

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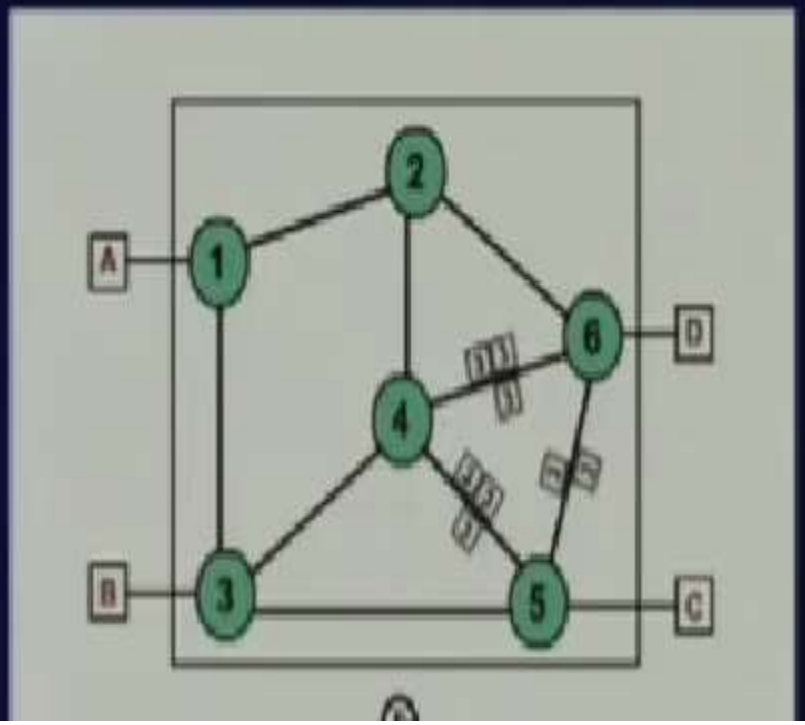

Flooding

- Requires no network information whatsoever
- Every incoming packet to a node is sent out on every outgoing line except the one it arrived on.
- Characteristics:
- All possible routes between source and destination are tried. A packet will always get through if a path exists.
- As all routes are tried, at least one packet will pass through the shortest route
- All nodes, directly or indirectly connected, are visited

- Limitation: Flooding generates vast number of duplicate packets
- Suitable damping mechanism must be used.

Flooding

- Hop-count: A hop counter may be contained in the packet header, which is decremented at each hop, with the packet being discarded when the counter becomes zero.
- The sender initializes the hop counter. If no estimate is known, it is set to the full diameter of the subnet. Worst case value
- Keep track of packets which are responsible for flooding using a sequence number. Avoid sending them out a second time.

Selective Flooding

- A variation which is slightly more practical is selective flooding.
  - The routers do not send every incoming packet out on every line, only on those lines that go in approximately in the direction of destination.

Flooding

- Utilities of flooding:
  - Flooding is highly robust, and could be used to send emergency messages (e.g., military applications).
  - May be used to initially set up the route in a virtual circuit.
  - Flooding always chooses the shortest path, since it explores every possible path in parallel.
  - Can be useful for the dissemination of important information to all nodes (e.g., routing information).

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Review Questions

1. Why routing is important in a packet-switched network?
2. What are the primary conditions that affect routing?
3. What is flooding? Why flooding technique is not commonly used for routing?
4. In what situation flooding is most appropriate? How the drawbacks of flooding can be minimized?

1. How the drawback of circuit-switching is overcome in message switching?

Ans: Message switching is based on store and forward technique. Instead of establishing a dedicated path, the message is sent to the nearest directly connected node. Each node stores the message, checks for error and forwards it. It allows more devices to share the network bandwidth and one message can be sent to several users. Destination host need not be on at the time of sending message.

2. What is the drawback of message switching? How is it overcome in packet switching?

Ans: In message switching, large storage space is required at each node to buffer the complete message blocks. On the other hand, in packet switching, messages are divided into subset of equal length, which are generated in the source node and reassembled to get back the initial complete message in destination node. Moreover, to transmit a message of large size, link is kept busy for a long time leading to increase in delay for other messages.

3. What are the key differences between datagram and virtual-circuit packet switching?

Ans: In datagram, the packets are routed independently and it might follows different routes to reach the destination in different order. In virtual-circuit packet switching, first a virtual connection is being established, and all the packets are sent serially through the same path. In this case, and packets are received in order.

4. Distinguish between circuit-switching and virtual-circuit packet switching.

Ans: In circuit switching, a dedicated path is established. Data transmission is fast and interactive. Nodes need not have storage facility. However, there is a call setup delay. In overload condition, it may block the call setup. It has fixed bandwidth from source to destination and no overhead after the call setup.

In virtual-circuit packet switching, there is no dedicated path. It requires storage facility and involves packet transmission delay. It can use different speed of transmission and encoding techniques at different segments of the route.

5. How packet size affects the transmission time in a packet switching network?

Ans: Initially, transmission time decreases as packet size is reduced. But, as packet size is reduced and the payload part of a packet becomes comparable to the control part, transmission time increases.